UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/027,167	09/21/2020	Yu Fang	SONA-0006-U01	7243

87084 7590 09/08/2021

GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
LIN, JSING FORNG	

ART UNIT	PAPER NUMBER
2446	

NOTIFICATION DATE	DELIVERY MODE
09/08/2021	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 17/027,167	Applicant(s) Fang et al.	
	Examiner JSING FORNG LIN	Art Unit 2446	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 9/221/2020.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-19 and 637 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-19 and 637 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

This Action is in response to application filed on 9/21/2020.

Claims 1-19, 637 are pending in this application.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 1/27/2021, 4/20/2021, 7/1/2021 are in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Double Patenting

A rejection based on double patenting of the "same invention" type finds its support in the language of 35 U.S.C. 101 which states that "whoever invents or discovers any new and useful process... may obtain a patent therefor..." (Emphasis added). Thus, the term "same invention," in this context, means an invention drawn to identical subject matter. See Miller v. Eagle Mfg. Co., 151 U.S. 186 (1894); In re Vogel, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and In re Ockert, 245 F.2d 467, 114 USPQ 330 (CCPA 1957).

A statutory type (35 U.S.C. 101) double patenting rejection can be overcome by canceling or amending the claims that are directed to the same invention so they are no

longer coextensive in scope. The filing of a terminal disclaimer cannot overcome a double patenting rejection based upon 35 U.S.C. 101.

Claims 1 are provisionally rejected under 35 U.S.C. 101 as claiming the same invention as that of claims 1 of copending Application No. 17,233,319. This is a provisional statutory double patenting rejection since the claims directed to the same invention have not in fact been patented.

Present Application No. 17,027,167	U.S. Application No. 17,233,319
Claim 1. A system comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured to interpret a policy comprising a network regulation description;	Claim 1. A system comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone; a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured to interpret a policy comprising a network regulation description;

a configuration circuit structured to configure at least one network interface circuit in response to the policy; and	a configuration circuit structured to configure at least one network interface circuit in response to the policy; and
the at least one network interface circuit, wherein the at least one network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone.	the at least one network interface circuit, wherein the at least one network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone.

CLAIM INTERPRETATION

The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

This application includes one or more claim limitations that use the word “means” or “step” or “structured to” but are nonetheless not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph because the claim limitation(s)

recite(s) sufficient structure, materials, or acts to entirely perform the recited function.

Such claim limitation(s) is/are:

- ... manager circuit structured to interpret ... claim 1 limitation 2
- ... configuration circuit structured to configure ... claim 1 limitation 3
- ... interface circuit structured to regulate ... claim 1 limitation 4

Examiner interpret circuit as hardware with specific structure corresponding to specification (for example paragraph 0347, 0363)

Because this/these claim limitation(s) is/are **not** being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are **not** being interpreted to cover only the corresponding structure, material, or acts described in the specification as performing the claimed function, and equivalents thereof.

If applicant intends to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to remove the structure, materials, or acts that performs the claimed function; or (2) present a sufficient showing that the claim limitation(s) does/do not recite sufficient structure, materials, or acts to perform the claimed function.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claims 1-5, 7-11, 13-19, 637 are rejected under 35 U.S.C. 103 as being unpatentable over SCHOCH (US 2019/0356574 A1) in view of Roese et al (US 2009/0187968 A1).

Regarding claim 1, SCHOCH teaches a system comprising: a vehicle having a first network zone and a second network zone of a different type than the first network zone (SCHOCH fig. 1 vehicle internal network with plurality security zones communicating with different message type; [0023][0032]);

a converged network device (CND) interposed between the first network zone and the second network zone, the CND comprising: a policy manager circuit structured a policy comprising a network regulation description (SCHOCH [0047][0049] a domain controller may provide a firewall of separating two security zones based on predefined security criterion [0023]);

a configuration circuit structured to configure at least one network interface circuit in response to the policy; and (SCHOCH [0023] a domain controller is configured to transmit message between two security zones if the messages meet criterion; claim 10);

the at least one network interface circuit, wherein the at least one network interface circuit is structured to regulate communications between end points of the first network zone and end points of the second network zone (SCHOCH [0023] control device on different security zone communicate each other if predefined criteria meet; [0028]);

However, SCHOCH does not explicitly teach a circuit structured to interpret a policy comprising a network regulation description; (Note: SCHOCH teaches a domain controller with a firewall is used to separate zone based on security criterion as recite above)

Roese teaches a circuit structured to interpret a policy comprising a network regulation description and to configure at least one network interface circuit in response to the policy (Roese fig. 3 item #306 determine dynamic polices; item #309 set polices; [0034] determine policies and set polices of the attached connection);

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH in view of Roese to interpret a policy comprising a network regulation description and to configure at least one network interface circuit in response to the policy.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform operation more efficiently, i.e. update policy dynamically (Roese [0034]).

Regarding claim 2, SCHOCH-Roese teaches the system of claim 1, wherein the policy manager circuit is further structured to receive a policy communication from an external device, and to interpret the policy by performing, in response to the policy communication, one of storing the policy or updating a stored policy (Roese [0031] acquire policy from remote policy server; [0034] saving policies);

Regarding claim 3, SCHOCH-Roese teaches the system of claim 2, wherein the external device is communicatively coupled to the policy manager circuit through at least one of the first network zone or the second network zone (SCHOCH [0039][0042] external server or smart phone couple to vehicle network domain controller);

Regarding claim 4, SCHOCH-Roese teaches the system of claim 2, wherein the external device is communicatively coupled to the policy manager circuit via at least one of a wireless network connection or a cellular network connection (SCHOCH [0039][0042] wireless connection);

Regarding claim 5, SCHOCH-Roese teaches the system of claim 2, wherein the policy manager circuit is further structured to verify the policy before performing the one of storing the policy or updating the stored policy (Roese [0033-0034] when the policy has been determined, save policy);

Regarding claim 7, SCHOCH-Roese teaches the system of claim 1, wherein the policy comprises at least one data collection parameter (Roese fig. 5 policy include port speed, bandwidth limit etc.; SCHOCH [0032]);

Regarding claim 8, SCHOCH-Roese teaches the system of claim 1, wherein the policy comprises a permission value corresponding to at least one end point of the first network zone or the second network zone, wherein the permission value comprises at least one permission value selected from the values consisting of: a data collection permission value; a service publication permission value; a service subscription permission value; or an external communication permission value (SCHOCH [0032] safety criterion include meet the predefined type spec);

Regarding claim 13, SCHOCH-Roese teaches the method of claim 12, wherein verifying the policy cannot be applied comprises at least one operation selected from the operations consisting of: determining that the policy violates a security criteria of a previously stored policy or the policy communication; determining that the policy cannot be implemented; determining that the policy cannot be implemented due to a network communication limit; determining that the policy cannot be implemented due to a data storage limit; or determining that the policy cannot be implemented due to an external communication limit (Roese [0029] determine additional network challenge passed? If not, deny network entry; SCHOCH [0032] safety criterion include meet the predefined type spec);

Regarding claim 14, SCHOCH-Roese teaches the method of claim 9, further comprising regulating external communications of end points of each of the first network zone and the second network zone in response to the policy (SCHOCH [0023] regulating between security zones);

Regarding claim 15, SCHOCH-Roese teaches the method of claim 9, further comprising regulating communications between a first end point of the first network zone and a second end point of the first network zone in response to the policy (SCHOCH [0023] regulating between control devices);

Regarding claim 16, SCHOCH-Roese teaches the method of claim 9, further comprising updating a local configuration file of a configurable ethernet switch in response to one of an implementation of the policy or an update of the policy (Roese fig. 1 item #104b, [0034] attached connection include switch);

Regarding claim 17, SCHOCH-Roese teaches the method of claim 9, further comprising updating a local configuration file of a configurable edge gateway in response to one of an implementation of the policy or an update of the policy (Roese fig. 1 item #102i [0034] attached connection include gateway);

Regarding claim 18, SCHOH-Roese teaches the method of claim 9, wherein regulating the communication between a first end point of the first network zone and a second end point of the second network zone comprises performing one of an up-

sampling operation or a down-sampling operation on the communication (Roese [0033][0040-0041] bandwidth setting dynamically based on dynamic policy);

Regarding claim 19, SCHOCH-Roese teaches the method of claim 9, wherein regulating the communication between a first end point of the first network zone and a second end point of the first network zone comprises performing one of an up-sampling operation or a down-sampling operation on the communication (Roese [0040-0041]);

Regarding claim 637, SCHOCH-Roese teaches the system of claim 1, further comprising wherein the policy is communicated to the CND from an external device. (SCHOCH [0039][0042], Roese [0031]);

Regarding claims 9, 10, 11, they do not teach or further define over the limitations in claims 1, 2, 5 respectively. Therefore, claims 9, 10, 11 are rejected for the same reasons as set forth in claims 1, 2, 5.

Claims 6, 12 are rejected under 35 U.S.C. 103 as being unpatentable over SCHOCH (US 2019/0356574 A1) in view of Roese et al (US 2009/0187968 A1) as applied to claims 1-5, 7-11, 13-19, 637 above and further in view of Penilla et al (US 2015/0210287 A1)

Regarding claim 6, SCHOCH-Roese teaches the system of claim 5,
As recited above.

However, SCHOCH-Roese does not explicitly teaches wherein the policy manager circuit is further structured to provide a notification to the external device in response to the verifying the policy (Note: Roese discloses [0023][0036] administrator monitor change triggers of attached connection and policy function 108 can be located at remote policy server).

Penilla teaches wherein the policy manager circuit is further structured to provide a notification to the external device in response to the verifying the policy (Penilla [0140] notification is triggered to the user's mobile device if violation in the restrictions are detected).

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify SCHOCH-Roese in view of to wherein the policy manager circuit is further structured to provide a notification to the external device in response to the verifying the policy.

One of ordinary skill in the art before the effective filing date of the claimed invention would have been motivated because it would have enabled the system to perform operation more efficiently (Penilla [0140]).

Regarding claim 12, SCHOCH-Roese-Penilla teaches the method of claim 11, further comprising providing a notification to the external device in response to the verifying the policy cannot be applied. (Penilla [0140], Roese [0029])

Additional References

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

a. McKeefery et al., US 20200252339 A1: SYSTEM, METHOD AND APPARATUS FOR RESTRICTING USE OF A NETWORK DEVICE THROUGH AUTOMATED POLICY ENFORCEMENT
b. Akella et al, US 20200036717 A1: SERIAL NETWORK COMMUNICATION USING INTELLIGENT ACCESS POLICIES which recite message exchange between CAN bus and Ethernet [0045];
c. KIM et al., US 20190394305 A1: APPARATUS FOR MANAGING SDN-BASED IN-VEHICLE NETWORK AND CONTROL METHOD THEREOF
d. SCHMIDT et al., US 20180270230 A1: MOTOR VEHICLE COMMUNICATION NETWORK WITH SWITCH DEVICE
e. Joshi et al., US 20200267080 A1: SYSTEMS AND METHODS FOR DATA PROCESSING AND STORAGE IN VEHICLES HAVING A ZONE-BASED, CENTRAL COMPUTING IN-VEHICLE COMMUNICATIONS NETWORK ARCHITECTURE
f. Shalev et al., US 20190260800 A1: CRYPTIC VEHICLE SHIELD
g. King et al., US 10650621 B1: Interfacing with a vehicular controller area network

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEVE LIN whose telephone number is 571-272-5137. The examiner can normally be reached on Monday – Friday 7:30 AM – 5:00 PM.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Brian Gillis can be reached on 571-272-7952. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/S. L./
Examiner, Art Unit 2446

/BRIAN J. GILLIS/
Supervisory Patent Examiner, Art Unit 2446